Project 1

Implementing CFS scheduling

Due date
2026. 04. 24. 23:59

Overview

- **Goal**
  - Replace xv6's default Round Robin scheduler with a simplified version of Linux's **Completely Fair Scheduler (CFS)**.
- **Submission Guidelines**
  - Submit the modified **xv6 source code** and the **report** for this assignment.
  - Push all your source code and your report to the following GitHub Classroom repository's **main branch**:
    - https://classroom.github.com/a/Z6qG6k3F
- **Deadline: April 24, 2026 (Wed) 23:59**
  - Late submissions are allowed, but your final score will be reduced by **5%** for each **3-hour window** following the due date.
  - Grading is based on your **final commit**. Be extremely cautious about committing after the deadline, as it will trigger the late penalty.

- There will be a **short quiz** right before the midterm exam on **April 25, 2026 (Sat)**.

DO NOT ASK THIS ISSUE AGAIN

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Repository Access Issue

You no longer have access to your assignment repository. Contact your teacher for support

Your assignment is due by **Apr 1, 2026, 14:59 UTC**

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Discussions

- ↑ 1 [Assignment 1] <레포지토리 접근 실패> — WONHO-JEONG-HANYANG asked yesterday in Questions · Closed · Unanswered — 2
- ↑ 1 [assignment1] Assignment 1 레포지토리에 접근할 수가 없습니다. — yenru0 asked yesterday in Questions · Closed · Unanswered — 2
- ↑ 1 [assignment1] <레포지토리 접근 실패> — kwon120 asked 2 days ago in Questions · Closed · Answered — 2
- ↑ 1 [assignment1] assignment1 repository Access Issue 발생으로 인한 접근 불가 문의 — EUNCH asked 2 days ago in Questions · Closed · Answered — 2
- ↑ 1 assignment repository 접근이 불가능합니다 — yeeunkim97 asked 4 days ago in Questions · Closed · Answered — 4

ESeokhwan last week Maintainer — edited

이와 같이 메일을 못 찾겠는 경우에는 다음 형식의 url에 접속하여 invitation을 받아보시길 바랍니다.

`https://github.com/2026-HYU-ELE3021/{과제 이름}-{GitHub username}/invitations`

예시) https://github.com/2026-HYU-ELE3021/assignment1-donghyeonahn/invitations

Unmark as answer · ↑ 1 · 1 reply

wnsah814 2 days ago Maintainer — edited

classroom 수락 후 메일로 보내진 초대 링크를 1주일 내로 수락하지 않으면 유효기간이 만료되어 접속이 불가능합니다.

다시 초대링크를 생성하였으니 확인해보시길 바랍니다.

https://github.com/2026-HYU-ELE3021/assignment1-kwon120/invitations

Unmark as answer · ↑ 1 · 1 reply

Core Concept of CFS

- **Completely Fair Scheduler (CFS)**
  - A scheduling algorithm that ensures CPU time is allocated proportionally among processes based on their priority.
  - It achieves this by consistently selecting and executing the runnable process with the lowest **virtual runtime**.
- **Virtual Runtime**
  - A metric that tracks the total CPU execution time a process has been scheduled, scaled by its priority.
  - Each process has **its own** virtual runtime. The scheduler updates the running process's virtual runtime based on its actual runtime and its priority.
- **Nice Value**
  - A priority hint that determines how rapidly virtual runtime is accumulated.
  - A **higher** nice value causes the scheduler to increase the virtual runtime value at a **faster** rate. As a result, the process is selected for execution **less frequently**.

Specifications

- **Process Selection**
  - Select the `RUNNABLE` process with the lowest virtual runtime.
  - Utilize the provided **Red-Black tree** data structure to manage runnable processes efficiently.

- **Red-Black Tree**
  - A self-balancing binary search tree that guarantees $O(\log n)$ time complexity for insertions, deletions, and lookups.

```
$ vim kernel/rbtree.h

struct rb_node {
  uint64 key;
  void *data;
  struct rb_node *_parent;
  struct rb_node *_left;
  struct rb_node *_right;
  rb_color _color;
};

struct rb_tree {
  struct rb_node *root;
};

void rb_insert(struct rb_tree *tree, struct rb_node *node);
void rb_delete(struct rb_tree *tree, struct rb_node *node);

struct rb_node* rb_first(struct rb_node *node); // minimum
struct rb_node* rb_last(struct rb_node *node); // maximum
```

Specifications

- **Virtual Runtime Initialization & Update**
  - Every time a process is selected, its virtual runtime is increased based on its priority.
  - **Each increment of 1** in the nice value **doubles** the amount of virtual runtime added.
  - When a process is **forked**, initialize its virtual runtime to **its parent's virtual runtime**. If there is no parent, initialize it to `0`.
  - When a process wakes up from `SLEEPING`, update its virtual runtime according to the following rules, evaluated in order:
    - If there are any `RUNNABLE` processes, set the waking process's virtual runtime to the minimum virtual runtime among all `RUNNABLE` processes.
    - If there are no `RUNNABLE` processes, but at least one process is currently `RUNNING`, set the waking process's virtual runtime to the lowest virtual runtime of any `RUNNING` process.
    - If there are neither `RUNNABLE` nor `RUNNING` processes, set the waking process's virtual runtime to 0.

Specifications

- **Nice value Initialization**
  - The nice value must always be in the range from `-3` to `2`.
  - Initialize the nice value to `0` (default priority) for the very first process.
  - When a process is forked, it inherits its parent's nice value.

- **`set_nice()` system call**
  - **Role:** Sets the nice value (priority) of the currently running process.
  - **Signature:**

```
int set_nice(int nice);
```

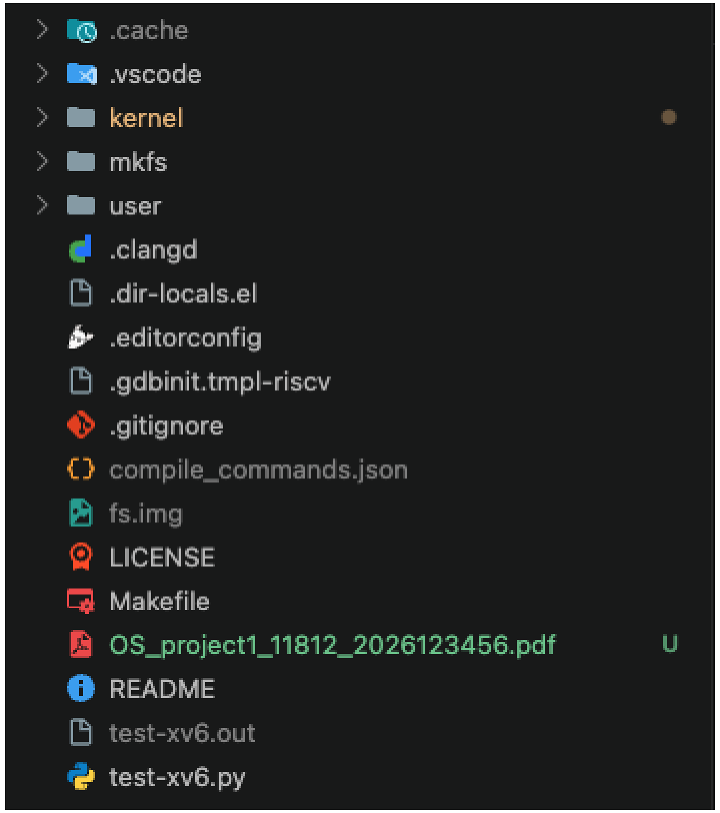
  - **nice (int):** The target nice value to apply. Must be within the valid range (from -3 to 2).
  - **Return value**:
    - Return 0 if the nice value is successfully updated.
    - Return 1 if the provided nice value is invalid.

Report

- **Design**
  - Explain your implementation plan to meet the assignment requirements.
- **Implementation**
  - Detail the specific modifications and additions you made to the given xv6 codebase. Clearly highlight how they differ from the original code and the purpose of these changes.
- **Results**
  - Document the compilation and execution process. Include screenshots verifying that all requirements are met, alongside explanations of the execution flow.
- **Troubleshooting**
  - If you encountered any issues during the assignment, describe the issues and your resolution process. For unresolved issues, explain what they were and what approaches you attempted.

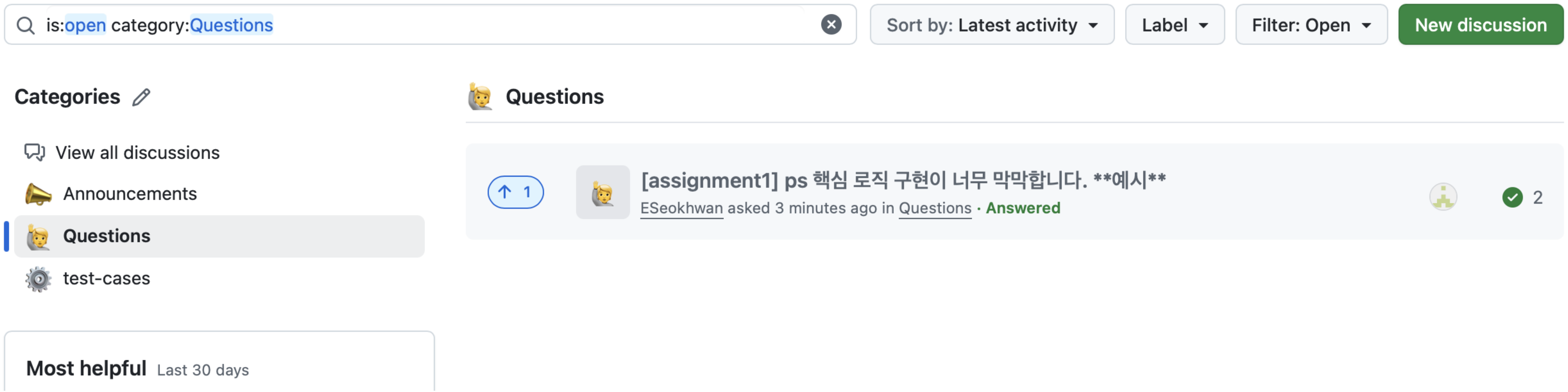
Report

- The report must be a PDF file strictly named: **OS_project1_[class number]_[student ID].pdf**
- You must commit and push your report PDF to your GitHub Classroom repository along with your xv6 source code.
- Make sure that report PDF is in the root directory of your repository, as shown in below.



Q&A

- For assignment related questions, please use the **Discussions tab** of xv6-sandbox repository.
  - https://github.com/2026-HYU-ELE3021/xv6-riscv-sandbox/discussions
- **Questions sent by email will not be answered.**
- For non-assignment questions or personal matters, please use an email like before.



- Students are highly **encouraged to answer each other's questions**! Active participation is greatly appreciated and helpful for everyone

Test Case Contribution

- You can share your custom test cases on the **Discussions tab** of xv6-sandbox repository too.
  - https://github.com/2026-HYU-ELE3021/xv6-riscv-sandbox/discussions
- A structured template is already provided, so we highly encourage your active contributions.
- Well-crafted test cases might actually be adopted for the official grading process!